Assignment #01

Department Of Economics

Subject Econometrics

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Example #2.6

Consider the following linear model:

$$Y_1 = \beta_1 + \beta_2 X_{2t} + \beta_3 X_{3t} + U$$

$t = 1, 2, 3, \ldots\ldots, T$

and the data in deviation form

$\sum y^2 = 390$ $\quad \sum x_2 y = 30$ $\quad \sum x_3 y = -20$

$\sum x_2 x_3 = 10$ $\quad \sum x_2^2 = 20$ $\quad \sum x_3^2 = 10$ $\quad T = 103$

(i) Compute β_2 and β_3

(ii) Compute R^2 and $\bar{R}^2$

Solution:- (i)

$$\hat{\beta}_2 = \frac{(\sum x_2 y)(\sum x_3^2) - (\sum x_3 y)(\sum x_2 x_3)}{(\sum x_2^2)(\sum x_3^2) - (\sum x_2 x_3)^2}$$

$$= \frac{(30)(10) - (-20)(10)}{(20)(10) - (10)^2}$$

$$= \frac{300 + 200}{200 - 100}$$

$$= \frac{5\cancel{00}}{1\cancel{00}}$$

$$\boxed{\hat{\beta}_2 = 5}$$

$$\hat{\beta}_3 = \frac{(\Sigma x_3 y)(\Sigma x_2^2) - (\Sigma x_2 y)(\Sigma x_2 x_3)}{(\Sigma x_2^2)(\Sigma x_3^2) - (\Sigma x_2 x_3)^2}$$

$$= \frac{(-20)(20) - (30)(10)}{(20)(10) - (10)^2}$$

$$= \frac{-400 - 300}{200 - 100} = \frac{-7\cancel{00}}{1\cancel{00}}$$

$$\boxed{\hat{\beta}_3 = -7}$$

(ii) R^2 and $\bar{R}^2$

$$R^2 = \frac{\hat{\beta}_2 \Sigma x_2 y + \hat{\beta}_3 \Sigma x_3 y}{\Sigma y^2}$$

$$= \frac{(5)(30) + (-7)(-20)}{390}$$

$$= \frac{150 + 140}{390} = \frac{290}{390}$$

$$\boxed{R^2 = 0.743}$$

$$\bar{R}^2 = 1 - (1 - R^2)\left(\frac{T-1}{T-k}\right)$$

$$= 1 - (1 - 0.743)\left(\frac{103-1}{103-3}\right)$$

$$= 1 - 0.257\left(\frac{102}{100}\right)$$

$$= 1 - 0.26214$$

$$\boxed{\bar{R}^2 = 0.737}$$

(b)

The following table shows the value of expenditure on cloth (Y) total expenditure (X) and price of clothing (X_2)

Year	1970	1971	1972	1973	1974	1975	1976
Y	6	7	9	8	10	12	14
X_1	42	50	54	65	72	85	90
X_2	7	7	5	4	3	3.5	2

Fit a least square regression equation of Y on X_1 and X_2 and gives the economic meaning of your estimate.

Solution:-

The desired multiple regression is

$$\hat{Y} = a + b_1X_1 + b_2X_2$$

Y	X_1	X_2	X_1^2	X_2^2	X_1X_2	X_1Y	X_2Y
6	42	7	1764	49	294	252	42
7	50	7	2500	49	350	350	49
9	54	5	2916	25	270	486	45
8	65	4	4225	16	260	520	32
10	72	3	5184	9	216	720	30
12	85	3.5	7225	12.25	297.5	1020	42
14	90	2	8100	4	180	1260	28
$\Sigma Y = 66$	$\Sigma X_1 = 458$	31.5	$\Sigma X_1^2 = 31914$	$\Sigma X_2^2 = 164.25$	$\Sigma X_1X_2 = 1867.5$	$\Sigma X_1Y = 4608$	$\Sigma X_2Y = 268$

Now we find

$\sum x_1^2 = \sum X_1^2 - \frac{(\sum X_1)^2}{n} = 31914 - \frac{(458)^2}{7} = 31914 - 29966.2857$

$\boxed{\sum x_1^2 = 1947.71}$

$\sum x_2^2 = \sum X_2^2 - \frac{(\sum X_2)^2}{n} = 164.25 - \frac{(31.5)^2}{7} = 164.25 - 141.75$

$\boxed{\sum x_2^2 = 22.5}$

$\sum x_1x_2 = \sum X_1X_2 - \frac{\sum X_1 \sum X_2}{n} = 1867.5 - \frac{(458)(31.5)}{7} = 1867.5 - 2061$

$\boxed{\sum x_1x_2 = -193.5}$

$\sum x_1y = \sum X_1Y - \frac{(\sum X_1)(\sum Y)}{n} = 4608 - \frac{(458)(66)}{7} = 4608 - 4318.2$

$\boxed{\sum x_1y = 289.71}$

$\sum x_2y = \sum X_2Y - \frac{(\sum X_2)(\sum Y)}{n} = 268 - \frac{(31.5)(66)}{7} = 268 - 297$

$\boxed{\sum x_2y = -29}$

Where

$$b_1 = \frac{(\sum x_1y)(\sum x_2^2) - (\sum x_2y)(\sum x_1x_2)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1x_2)^2}$$

$$= \frac{(289.71)(22.5) - (-29)(-193.5)}{(1947.71)(22.5) - (-193.5)^2}$$

$$= \frac{6518.475 - 5611.5}{43823.475 - 37442.25} = \frac{906.75}{6321}$$

$\boxed{b_1 = 0.142}$

$$b_2 = \frac{(\sum x_2y)(\sum x_1^2) - (\sum x_1y)(\sum x_1x_2)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1x_2)^2}$$

$$= \frac{(-29)(1947.71) - (289.7)(-193.5)}{(1947.71)(22.5) - (-193.5)^2}$$

$$= \frac{-56483.3 + 56056.95}{43823.475 - 37442.25} = \frac{-426.35}{6321}$$

$\boxed{b_2 = -0.067}$

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and $a = \bar{Y} - b_1\bar{X}_1 - b_2\bar{X}_2$

$\bar{Y} = \frac{\Sigma Y}{n} = \frac{66}{7}$; $\bar{X}_1 = \frac{X_1}{n} = \frac{458}{7}$; $\bar{X}_2 = \frac{\Sigma X_2}{n} = \frac{31.5}{7}$

$\boxed{\bar{Y} = 9.429}$ $\boxed{\bar{X}_1 = 65.429}$ $\boxed{\bar{X}_2 = 4.5}$

Now $a = 9.429 - (0.142)(65.429) - (0.067)(4.5)$

$= 9.429 - 9.291 + 0.302$

$\boxed{a = 0.440}$

Hence the fitted least square regression equation of Y on X_1 and X_2 is

$\hat{Y} = 0.440 + 0.142X - 0.067X_2$

(C)

The following are data on Y: the average weekly profit in (thousand-rupees) of five restaurants X_1: their seating capacities, and X_2 the average daily traffic.

X_1	120	200	150	180	240
X_2	19	08	12	25	16
Y	23.8	24.2	22.0	26.2	33.5

Fit an equation of the form

$Y = b_0 + b_1X_1 + b_2X_2$ ~~an~~

and find $R^2_{y.12}$ coefficient of determintion

Solution

The multiple least square regression line is

$$Y = b_0 + b_1 X_1 + b_2 X_2$$

where regression coefficients are b_1, b_2 and regression constant is

$$a = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2 .$$

Y	X_1	X_2	X_1^2	X_2^2	X_1Y	X_2Y	X_1X_2
23.8	120	19	14400	361	2856	452.2	2280
24.2	200	8	40000	64	4840	193.6	1600
22.0	150	12	22500	144	3300	264	1800
26.2	180	15	32400	225	4716	393	2700
33.5	240	16	57600	256	8040	536	3840
129.7	890	70	166900	1050	23752	1838.8	12220

Now we calculate

$$\sum x_1^2 = \sum X_1^2 - \frac{(\sum X_1)^2}{n} = 166900 - \frac{(890)^2}{5}$$

$$\boxed{\sum x_1^2 = 8480}$$

$$\sum x_2^2 = \sum X_2^2 - \frac{(\sum X_2)^2}{n} = 1050 - \frac{(70)^2}{5}$$

$$\boxed{\sum x_2^2 = 70}$$

$$\sum x_1 x_2 = \sum X_1 X_2 - \frac{(\sum X_1)(\sum X_2)}{n} = 12220 - \frac{(890)(70)}{5}$$

$$\boxed{\sum x_1 x_2 = -240}$$

$$\sum x_1 y = \sum X_1 Y - \frac{(\sum X_1)(\sum Y)}{n} = 23752 - \frac{(890)(129.7)}{5}$$

$$\boxed{\sum x_1 y = 665.4}$$

$$\sum x_2 y = \sum X_2 Y - \frac{(\sum X_2)(\sum Y)}{n} = 1838.8 - \frac{(70)(129.7)}{5}$$

$$\boxed{\sum x_2 y = 23}$$

Where

$$b_1 = \frac{(\sum x_1 y)(\sum x_2^2) - (\sum x_2 y)(\sum x_1 x_2)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$= \frac{(665.4)(70) - (23)(240)}{(8480)(70) - (-240)^2}$$

$$= \frac{46578 + 5520}{593600 - 57600} = \frac{52098}{536000}$$

$$\boxed{b_1 = 0.972}$$

$$b_2 = \frac{(\sum x_2 y)(\sum x_1^2) - (\sum x_1 y)(\sum x_1 x_2)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$= \frac{(23)(8480) - (665.4)(-240)}{(8480)(70) - (-240)^2}$$

$$\boxed{b_2 = 0.6618}$$

Now we find constants w.r.t $\bar{Y}, \bar{X}_1, \bar{X}_2$, b_1 and b_2.

$\bar{Y} = \frac{\sum Y}{n} = 25.94$ $\quad \bar{X}_1 = \frac{\sum X_1}{n} = 178$, $\bar{X}_2 = \frac{\sum X_2}{n} = 14$

$$a = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2$$

$$a = 25.94 - 0.0972(178) - 0.6618(14)$$

$$\boxed{a = -0.6268}$$

Hence the desired multiple linear regression equation is

$$\boxed{Y = -0.6268 + 0.0972 X_1 + 0.6618 X_2}$$

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$$R^2_{y12} = \frac{a\sum Y + b_1 \sum X_1 Y + b_2 \sum X_2 Y - (\sum Y)^2/n}{\sum Y^2 - (\sum Y)^2/n}$$

$$= \frac{-0.6268(1297)+0.972(23752)+0.6618(1888.8)-3[illegible]}{3444.77-3364.418}$$

$$= \frac{79.89824}{80.352}$$

$$\boxed{R^2_{y12} = 0.994}$$

Question

An appliance store conducts a five month experiment to determine the effect of advertising ~~on sales revenue~~

Advertising Expenditure, x (\$100s)	1	2	3	4	5
Sales Revenue, Y (\$1000s)	1	1	2	2	4

The following regression is estimated by OLS

$$\hat{Y} = -0.1 + 0.7x$$

Find the elasticity of sales revenue and interpret.

Solution

$$E_{Y/X} = \frac{\partial Y}{\partial X} \frac{\bar{X}}{\bar{Y}} \quad \text{(Average elasticity)}$$

$$= (0.7)\left(\frac{3}{2}\right)$$

$$= 2.1$$

$$\boxed{E_{Y/X} = 1.05}$$

Now we find

5 Interpret:- The elasticity coefficient, 1.05 shows that around the mean values of the variables, a 1-precent in advertising expenditure, x will lead to 1.05 percent increase in sales revenue Y.

(b)

Given the following data of quantity demanded (Y) and Price (X).

X	3	4	5	6	7	8	9	10	11	12
Y	25	24	20	20	19	17	16	13	10	6

Estimate the linear function by OLS, interpret the slope coefficient and compute price elasticity of demand at price = 8

Solution

$n = 10$, $\Sigma X = 75$, $\Sigma X^2 = 645$, $\Sigma Y = 170$,
$\Sigma XY = 1116$, $\hat{\alpha} = 31.454$, $\hat{\beta} = -1.927$

Thus

$$\hat{Y} = 31.454 - 1.927X$$

$\hat{\beta} = -1.927$ gives that for every 1 unit increase in price the quantity demand will decrease by 1.927 units and vice versa.

Now

$$Y_0 = 31.454 - 1.927(8)$$

$$\boxed{Y_0 = 16.038}$$

And

$E = \hat{\beta}\left(\frac{X_0}{Y_0}\right)$ Point elasticity

$E = -1.927\left(\frac{8}{16.038}\right)$

$\boxed{E = -0.96}$

The elasticity coefficient tell that around the specific value (8 and 16.038) of the variables, a 1-percent increase in price will lead to 0.96 percent decrease in quantity demanded.

Question

The demand for steel of a major product has been estimated as.

$$\hat{Y}_t = 66{,}000 - 1{,}100X_{2t} + 0.2\,X_{3t} + 105X_{4t}$$

Where $\hat{Y}$ is the estimate quantity demanded for steel, X_2 is the price of steel, X_3 is the per capita income, X_4 is the price of substitute. If typical price of steel is Rs.12/Lb, per capita income is Rs.3000 and price of aluminium is Rs.10/LB. Find and interpret.

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Solution

$$Y_0 = 60{,}000 - 1{,}200(12) + 0.2(3000) + 105(10)$$

$$\boxed{Y_0 = 48450}$$

(a) Own Price elasticity of demand for steel.

$$E_{Y/X_2} = \frac{\partial Y}{\partial X_2}\left(\frac{X_{0(2)}}{Y_0}\right)$$

$$= 1{,}100\left(\frac{12}{48450}\right)$$

$$E_{Y/X_2} = -0.272$$

The elasticity coefficient (−0.272) tells that around the specific values of variables ($X_{0(2)}$, Y), a 1-percent increase in Price of steel will lead to decrease 0.272 percent demand for steel

(b) Cross Price elasticity of demand for steel with aluminium.

$$E_{Y/X_4} = \frac{\partial Y}{\partial X}\left(\frac{X_{0(4)}}{Y_0}\right)$$

$$= (105)\left(\frac{10}{48450}\right)$$

$$= \frac{1050}{48450}$$

$$\boxed{E_{Y/X_4} = 0.022}$$

It means around the given values of variables, a 1-percent change in price of substitute will lead to 0.022 percent change in demand for steel.

(C) Income elasticity of demand for steel.

$$E_{Y/X_3} = \frac{\partial Y}{\partial X_3}\left(\frac{X_{0(3)}}{Y_0}\right)$$

$$= (0.2)\left(\frac{3000}{48450}\right) = \frac{600}{48450}$$

$$\boxed{E_{Y/X_3} = 0.0124}$$

It means around the specific values of the variables, a 1 percent change in per capita income will lead to 0.012 percent change in demand for steel.

(d) the variables, to which, demand for steel is most sensitive.

The demand for steel is most sensitive with price of steel, since the absolute value of elasticity coefficient for price is greater than from the absolute elasticity coefficient, for price of substitute & per capita income.

Question

Consider the following regression model

(C)

(i) $Y_i = \alpha_1 + \alpha_2 X_{2i} + \varepsilon_i$

(ii) $\ln Y_i = \beta_1 + \beta_2 \ln X_{2i} + \varepsilon_i$

(iii) $Y_i = \gamma_1 + \gamma_2 \ln X_{2i} + \varepsilon_i$

(iv) $\ln Y_i = \delta_1 + \delta_2 X_{2i} + \varepsilon_i$

(a) Interpret $\alpha_2, \beta_2, \gamma_2$ and δ_2

(i) The one unit change in X_2 will bring a particular magnitude, α_2 unit change in Y. In other words, the slop coefficient α_2 measure the absolute change in Y for a given absolute change in X.

$$\left(\alpha_2 = \frac{\text{Absolute change in } Y}{\text{Absolute Change in } X}\right)$$

(ii) The slope coefficient β_2 measures the elasticity of Y with respect to X_2, that is, the percentage change in Y for a given % change in X_2.

$$\left(\beta_2 = \frac{\text{Relative Change in } Y}{\text{Relative Change in } X} = \frac{d(\ln Y)}{d(\ln X)}\right)$$

Note that, the change in $\ln Y$ is equal to relative or proportional change in Y.

(iii) The slope coefficient γ_2 measures the absolute change in Y for a relative change in X. If relative change in X is multiplied by 100 then γ_2 measures the absolute change in Y for a % change in X. Thus, the slop coefficient, means that an increase in X of 1% on average, leads to about $\frac{\gamma_2}{100} = 0.01\gamma_2$ units increase in Y.

(iv) The slop coefficient δ_2 measures the constant proportional or relative change in Y for a given absolute change in X. By multiplied relative change in Y by 100, δ_2 gives the ~~perent~~ % change, or growth rate in Y, for an absolute change in X. that is $100 \times \delta_2$ gives the growth rate in Y.

(b)

(i) $\eta = (\gamma_2)\frac{\bar{X}}{\bar{Y}}$ around the mean values of the variables, a 1 percent increase in X will lead to "η"% increase in Y, the dependent variable.

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(ii) $\eta = \beta_2$ (constant elasticity)

that is, the change in $\ln Y$ per unit change in $\ln X$ remains the same no matter at which $\ln X$ we measure the elasticity. β_2 is the elasticity coefficient.

(iii) $$\eta = \frac{\partial Y}{\partial X_{2i}} \cdot \frac{X_{2i}}{Y_i}$$

Then $$\frac{\partial Y_i}{\partial X_{2i}} = \frac{\gamma_2}{X_{2i}}$$

Therefore

$$\eta = \left(\frac{\gamma_2}{X_{2i}}\right)\left(\frac{X_{2i}}{Y}\right) = \frac{\gamma_2}{Y_i}$$

that is elasticity is not constant and changes inversely with Y_i.

(iv) $$\frac{\partial(\ln Y_i)}{\partial X_{2i}} = \frac{1}{Y}\frac{\partial Y_i}{\partial X_{2i}} = \delta_2 \Rightarrow \frac{\partial Y}{\partial X} = \delta_2 Y$$

$$\eta = \frac{\partial Y_i}{\partial X_{2i}}\left(\frac{X_{2i}}{Y_i}\right) = (\delta_2 Y_i)\left(\frac{X_{2i}}{\cancel{Y_i}}\right)$$

$$\boxed{\eta = \delta_2 X_{2i}}$$